CHAPTER 7

# GRAVITATION

**CHAPTER ESSENCE AT A GLANCE:**
Gravity reaches across empty space! Every mass attracts every other mass with an inverse-square force. Launch a satellite fast enough (~7.9 km/s) and it falls around the Earth without ever hitting it; launch at 11.2 km/s and it escapes Earth forever!

## 4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Universal Gravitation:** F = G M m divided by $r^2$ where G = 6.67 × $10^{-11}$ N $m^2$ $kg^{-2}$. Acceleration due to gravity at surface: g = G M divided by $R^2$ approx. 9.8 $m/s^2$.

• **Variation of g:** With Altitude h: g h approx. g(1 - 2h divided by R) for h << R. With Depth d: g d = g(1 - d divided by R). At Earth's center (d = R), g = 0! At height h = d divided by 2, gravity matches depth d.

• **Orbital vs Escape Velocity:** Orbital speed near Earth $v_0$ = sqrt(gR) approx. 7.92 km/s. Escape speed v e = sqrt(2gR) = sqrt(2) $v_0$ approx. 11.2 km/s. Escape speed is independent of the projection angle!

• **Why Moon Has No Atmosphere:** Due to small mass and radius, Moon's escape speed is only v e approx. 2.38 km/s. Gas thermal speed v rms exceeds 2.38 km/s (v rms > v e), so gases escaped into space.

## ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$F = G\frac{Mm}{r^2}, \quad g = \frac{GM}{R^2}$$

*Universal Gravitation and Acceleration at Surface*

$$g_h = g\left(1 - \frac{2h}{R}\right), \quad g_d = g\left(1 - \frac{d}{R}\right)$$

*Variation of g with Altitude and Depth*

$$U(r) = -\frac{GMm}{r}, \quad V(r) = -\frac{GM}{r}$$

*Gravitational Potential Energy and Potential*

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}, \quad v_o = \sqrt{\frac{GM}{R}} = \sqrt{gR}$$

*Escape Speed and Orbital Satellite Speed*

## CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Escape velocity does NOT depend on the mass of the projectile! A feather and a rocket have the exact same escape speed from Earth.

• **Alert:** Remember Kepler's 3rd Law: $T^2$ is proportional to $R^3$. If radius quadruples, orbital period increases by a factor of 8 (4^(3 divided by 2) = 8)!

## HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

### 1. State Kepler's three laws of planetary motion. [Score: 2 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: First and Second Laws]** 1. Law of Orbits: All planets move in elliptical orbits with the Sun at one focus. 2. Law of Areas: The line joining a planet and the Sun sweeps out equal areas in equal intervals of time (dA divided by dt = L divided by (2m) = constant). [1 Mark]

• **[Step 2: Third Law]** 3. Law of Periods: The square of the orbital period of a planet is directly proportional to the cube of the semi-major axis of its orbit ($T^2 \propto a^3$). [1 Mark]

• **Key Exam Result:** 1. Ellipses, 2. dA divided by dt = const, 3. $T^2 \propto a^3$

### 2. Derive the relation between acceleration due to gravity 'g' and the Universal Gravitational Constant 'G'. [Score: 2 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Force Equation]** Consider a body of mass m on Earth's surface (mass M, radius R). Gravitational force F = G M m divided by $R^2$. [1 Mark]

• **[Step 2: Equating to Weight]** Weight of body W = m g. Since F = W: m g = G M m divided by $R^2$ => g = G M divided by $R^2$. [1 Mark]

• **Key Exam Result:** g = G M divided by $R^2$

### 3. Derive an expression for the variation of acceleration due to gravity with altitude h above the surface of the Earth. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Formula at Height h]** On surface: g = G M divided by $R^2$. At height h above surface: g h = G M divided by $(R + h)^2$ = G M divided by [$R^2$ (1 + h divided by R)$^2$] = g (1 + h divided by R)$^{-2}$. [1.5 Marks]

• **[Step 2: Binomial Approximation for h << R]** Expanding binomially for h << R and neglecting higher order terms: g h approx. g (1 - 2h divided by R). Acceleration due to gravity decreases with increasing altitude. [1.5 Marks]

• **Key Exam Result:** g h = g (1 - 2h divided by R) for h << R

### 4. Derive an expression for the variation of acceleration due to gravity with depth d below the surface of the Earth. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Earth Density & Mass at Depth d]** Assuming uniform density ρ: g = G M divided by $R^2$ = G [4 divided by 3 π $R^3$ ρ] divided by $R^2$ = 4 divided by 3 π G ρ R ---(1). At depth d below surface, effective radius is (R - d). Effective mass M' = 4 divided by 3 π $(R - d)^3$ ρ. [1.5 Marks]

• **[Step 2: Expression for g d]** g d = G M' divided by $(R - d)^2$ = 4 divided by 3 π G ρ (R - d) ---(2). Dividing (2) by (1): g d divided by g = (R - d) divided by R = 1 - d divided by R ⇒ g d = g (1 - d divided by R). At Earth's centre (d = R), g = 0. [1.5 Marks]

• **Key Exam Result:** g d = g (1 - d divided by R); at center g = 0

### 5. Define Gravitational Potential Energy. Derive an expression for the gravitational potential energy of a mass m placed at distance r from the centre of the Earth. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Definition]** Gravitational potential energy of a body at a point is the work done in bringing the body from ∞ to that point against the gravitational force: U = -W. [1 Mark]

• **[Step 2: Work Integration]** Gravitational force at distance x: F = G M m divided by $x^2$. Work done in moving through dx towards Earth: dW = F dx = (G M m divided by $x^2$) dx. [1 Mark]

• **[Step 3: Total Potential Energy]** W = $\int_{\infty}^{r}$ (G M m divided by $x^2$) dx = G M m [-1 divided by x] $\infty^{r}$ = - G M m divided by r. Gravitational potential energy U = - G M m divided by r. [1 Mark]

• **Key Exam Result:** U(r) = - G M m divided by r

**6. At what height above the Earth's surface does the acceleration due to gravity become one-fourth (0.25)th of its value on the surface? (Radius of Earth R = 6400 km) [Score: 3 Marks]**

**Answer & Key Valuation Steps:**

• **[Step 1: Formula Setup]** g h = g [R divided by (R + h)]$^2$. Given g h = g divided by 4. [1 Mark]

• **[Step 2: Algebraic Evaluation]** g divided by 4 = g [R divided by (R + h)]$^2$ ⇒ [R divided by (R + h)]$^2$ = one-fourth (0.25). Taking square root: R divided by (R + h) = half (0.5) ⇒ R + h = 2 R ⇒ h = R. [1.5 Marks]

• **[Step 3: Numerical Value]** h = R = 6400 km. [0.5 Mark]

• **Key Exam Result:** Height h = R = 6400 km

**7. Derive expressions for: (a) Orbital speed v o = sqrt(g R) of a satellite orbiting close to Earth's surface (b) Time period of the satellite T = 2 π sqrt(R divided by g). [Score: 4 Marks]**

**Answer & Key Valuation Steps:**

• **[Step 1: Orbital Speed Derivation]** For a satellite of mass m in circular orbit of radius r = (R + h) around Earth (mass M): Centripetal force = Gravitational force ⇒ m v o$^2$ divided by (R + h) = G M m divided by (R + h)$^2$ ⇒ v o = √[G M divided by (R + h)]. For orbit close to surface (h << R): v o = √(G M divided by R) = √(g R) (≈ 7.92 km/s). [2 Marks]

• **[Step 2: Time Period Derivation]** Orbital circumference = 2 π (R + h). Time period T = 2 π (R + h) divided by v o = 2 π (R + h) divided by √[G M divided by (R + h)] = 2 π √[(R + h)$^3$ divided by (G M)]. For orbit close to surface (h << R): T = 2 π √(R$^3$ divided by (G M)) = 2 π √(R divided by g) ≈ 84.6 minutes. [2 Marks]

• **Key Exam Result:** v o = √(g R) ≈ 7.9 km/s, T = 2 π √(R divided by g) ≈ 84.6 min